

ANSWER SCHEME PREPSPM DC014 KMNS 2024/2025

NO.	ANSWER	MARKS						
1(a)	• GRAPHIC - A digital representation of non-text information such as a drawing, chart or photo • AUDIO - The use of speech, music and sound effects produced by vibration, as perceived by the sense of hearing. • VIDEO - A series of frames containing pictures and sound played back at a fast rate. OR Video is a product of recording moving visual images by using a camera or video machine • ANIMATION - A series of graphical images displayed one after another to simulate movement. Or Animation is a process of making a static image look like it is moving.	Choose any one J2 (no mark for only element)						
1(b)	<table><tr><th>Scenario</th><th>Purpose of Multimedia Element</th></tr><tr><td>Les' Copaque Production brings the characters of the talking mouse deer, crocodile, and tiger to life in the story "Pada Zaman Dahulu."</td><td>Animation is useful for illustrating concepts which involve movement. or Inject humour and light-heartedness in a presentation via an animated cartoon. (any acceptable answer)</td></tr><tr><td>Police investigating officers try to draw the face of a criminal suspect based on the victim's testimony using an iPad</td><td>Graphic make complex information easy to understand. or Convey information more quickly than when using text. (any acceptable answer)</td></tr></table>	Scenario	Purpose of Multimedia Element	Les' Copaque Production brings the characters of the talking mouse deer, crocodile, and tiger to life in the story "Pada Zaman Dahulu."	Animation is useful for illustrating concepts which involve movement. or Inject humour and light-heartedness in a presentation via an animated cartoon. (any acceptable answer)	Police investigating officers try to draw the face of a criminal suspect based on the victim's testimony using an iPad	Graphic make complex information easy to understand. or Convey information more quickly than when using text. (any acceptable answer)	J1 J1
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1(c)	J : X-axis and Y-axis or X,Y S : X-axis , Y-axis and Z-axis or X,Y,Z	J1 J1						
2(a)(i)	Post Production	J1						
2(a)(ii)	Process after the production of a multimedia project.	J2						
2(a)(iii)	• Testing • Mastering • Archiving • Duplication • Distribution	Choose any one J1						

2(b)(i)	<table><tr><th>Authoring Tools</th><th>Characteristic</th><th>Example Software</th></tr><tr><td>Icon-based</td><td>The multimedia elements or events are organized as objects in a structural framework</td><td>Scratch, Icon Author, Macromedia Authorware</td></tr><tr><td>Page/Card-based</td><td>The multimedia elements or events are organized as pages of a book or a stack of cards</td><td>Google Slide, Prezi, Ms PowerPoint, ToolBook, HyperNext Studio, Hyper Studio, PhytionCard.</td></tr></table>	Authoring Tools	Characteristic	Example Software	Icon-based	The multimedia elements or events are organized as objects in a structural framework	Scratch, Icon Author, Macromedia Authorware	Page/Card-based	The multimedia elements or events are organized as pages of a book or a stack of cards	Google Slide, Prezi, Ms PowerPoint, ToolBook, HyperNext Studio, Hyper Studio, PhytionCard.	J2 For each characteristic J1 for each example			
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2(b)(ii)	Hand Gesture Control and handling multimedia applications through hand gesture recognition scheme	J1 J1												
2(c)	<table><tr><th>Description</th><th>Medium of Delivery</th><th>Example</th></tr><tr><td>A teacher uploads lessons and quizzes to an e-learning platform for students to access from home.</td><td>Online</td><td> • Web-based • Cloud-based Choose any one</td></tr><tr><td>A trainer gives participants a USB drive with all course materials to use.</td><td>Offline</td><td> • hard disk • solid state drive (SSD) • CD/DVD/Blu-Ray. Choose any one</td></tr></table>	Description	Medium of Delivery	Example	A teacher uploads lessons and quizzes to an e-learning platform for students to access from home.	Online	• Web-based • Cloud-based Choose any one	A trainer gives participants a USB drive with all course materials to use.	Offline	• hard disk • solid state drive (SSD) • CD/DVD/Blu-Ray. Choose any one	J1 (each medium) J1 (each example)			
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3(a)	<table><tr><th>Database System</th><th>File Processing System</th></tr><tr><td>All departments or areas within an organization share the data in the database.</td><td>Each department or area within an organization has its own set of files.</td></tr><tr><td>Databases are designed to store and manipulate large collections of organized information</td><td>Originally designed to store and manipulate data for a single user.</td></tr><tr><td>Accessible by allowing multiple users at the same time to quickly and securely access and query the data using highly complex logic and language.</td><td>Accessible by a single user or small number of users who don't need to do a lot of incredibly complicated data manipulation.</td></tr><tr><td>Relatively larger amount of data</td><td>Relatively smaller amount of data</td></tr><tr><td>Data Consistency – Data is only entered one time and stored in only one file (no duplication of data). So, there is no different versions of the same data.</td><td>Data Inconsistency – When different versions of the same data appear in different places.</td></tr></table>	Database System	File Processing System	All departments or areas within an organization share the data in the database.	Each department or area within an organization has its own set of files.	Databases are designed to store and manipulate large collections of organized information	Originally designed to store and manipulate data for a single user.	Accessible by allowing multiple users at the same time to quickly and securely access and query the data using highly complex logic and language.	Accessible by a single user or small number of users who don't need to do a lot of incredibly complicated data manipulation.	Relatively larger amount of data	Relatively smaller amount of data	Data Consistency – Data is only entered one time and stored in only one file (no duplication of data). So, there is no different versions of the same data.	Data Inconsistency – When different versions of the same data appear in different places.	Choose any one J2, J2
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	Sharing of Data – Data is shared over a network, by the entire organization. Limited Data Sharing - It is difficult to access data stored in separate files in different departments.	
3(b)	• Shared data – data is shared over a network, by the entire organization. • Reduced data redundancy – data is only entered one time and stored in only one file (no duplication of data). • Backup dan Recovery – automatically takes care of backup and recovery. • Improved data security – data is not accessible to unauthorized users.	Choose any one J2
4(a)	• Human error: When individuals enter information incorrectly, duplicate or delete data, don't follow the appropriate protocols, or make mistakes during the implementation of procedures. • Transfer error: Transfer errors happen when a piece of data is present in the destination table, but not in the source table in a relational database. • Bugs and viruses: Spyware, malware, and viruses are pieces of software that can invade a computer and alter, delete, or steal data. • Compromised hardware: Sudden computer or server crashes, and problems with how a computer or other device functions.	Choose any one J2
4(b)	• Limiting access to data - Limit and change permissions to restrict information by unauthorized parties. • Validating data - to make sure it is correct when it's used. • Backing up data - Save the data to another location storage. • Using logs to keep track of when data is added, modified, or deleted. • Conducting regular internal audits. • Using error detection software.	Choose any one J2
5(a)(i)	Guided: use wire, cable and other tangible materials to send communication signals Unguided: send communication signals through air or space	J2 J2
5(a)(ii)	Unguided media / Wireless/ Bounded media Flexibility: Unguided media, such as Wi-Fi and radio frequency communication, allows for easy setup and teardown, which is ideal for temporary events. There is no need to install permanent cabling, making it easier to adapt to the layout of the event space. Mobility: Attendees can move freely around the venue with mobile devices connected to a wireless network, facilitating communication and access to resources without being tethered to specific locations. Quick Deployment: Setting up a wireless network is generally faster than running cables, which is crucial when preparing for an event with a limited timeframe	J1 J2 Choose any one

	Or any related answers	
5(a)(iii)	- Cellular radio transmission - Microwave and satellite transmission - Infrared transmission	J1 J1 Choose any two
5(b)(i)	Fiber Optic Cable	J1
5(b)(ii)	Consists of dozens or hundreds of thin strands of glass or plastic that use light to transmit signals	J2
5(c)(i)	X: Router Y: Switch/Hub Z: Modem	J1 J1 J1
5(c)(ii)	A central communication device that allows computers and devices to transfer data wirelessly among themselves or to a wired network using wireless technologies.	J2
5(c)(iii)	NIC/ Network Interface Card/Network Card	J1
6(a)(i)	Search engine	J1
6(a)(ii)	Search engine is a software that find information on the web related to a specific topic.	J2
6(b)(i)	Using phrase	J1
6(b)(ii)	Search for exact words or phrases to narrow down the results even more.	J2
6(b)(iii)	“Computer Science” Or any related answers	J1
6(c)	Zara is researching how social media affects mental health and is looking for articles on "depression" or "anxiety." Boolean operators	J1
	Kumar is searching for recent reports on climate change policies in Negeri Sembilan from government websites. Field Search	J1
6(d)(i)	- an online platform which people use to build social networks - social relationships with other people who share similar personal - career content, interests, activities, backgrounds - real-life connections.	Choose any one J2
6(d)(ii)	- Instant Messaging /Text Messaging - online news, reference, and information - e-commerce - Online entertainment - online education and writing Or any related answers	Choose any one J1

6(d)(iii)	• Instant Messaging /Text Messaging ○ AIM, Yahoo! Messenger, WhatsApp, Telegram, Wechat ○ e-mail, SMS • Online news, reference, and information ○ BH ONLINE, myMetro, astro AWANI, CNN • e-commerce ○ Shopee, Lazada, Maybank2u, iRakyat, eBay, lelong.my • Online entertainment ○ Spotify, Apple Music, Amazon Music Unlimited, Netflix, Hulu, Amazon Prime Video • Online education and writing ○ Google classroom, Google meet, zoom, quizizz, Kahoot! Google Scholar	Choose any one J1
6(e)(i)	• Educate Users on Phishing Awareness • Be cautious when clicking on links or downloading attachments from unknown sources. Or any related answers	Choose any one J2
6(e)(ii)	• Prevent identity theft – Prevent/avoid Identity being used by someone that pretends to be you online. • Protect financial information - Cybercriminals can take our banking information and make unauthorized withdrawals, purchases, and transfers. • Protect your employability - To avoid damaging your online reputation, and thus sabotaging your career, you should remove any personal information that might negatively influence a prospective employer	Choose any one J2
6(f)(i)	• Home automation system - Enable the appliances and other devices in your home such as alarm system, door locks, and sprinklers : operate according to your stated preferences and can be controlled remotely via smartphone. • Fitness systems - Uses a sports-tested accelerometer to track your daily activity including running, walking, dancing etc in steps, time of day, and calories burned. You can set a personal goal for yourself and share your data with others via Facebook and Twitter. • Smart home system - Long-term benefits could include energy savings by automatically ensuring lighting, heating and air conditioning are turned off or by making the residents in the home aware of usage. • Elder care system - IoT can provide assistance to elderly individuals and to those with disabilities by the use of assistive technology such as voice control that can assist users with sight and mobility limitations while alert systems can be connected directly to cochlear implants worn by hearing-impaired users. Any acceptable answers	Choose any one J2

6(f)(ii)	• Chatbot • Google AI-Powered prediction • Facebook Face • Alexa	Choose any one J1
	TOTAL MARKS	70